|INCH-POUND|
MIL-I-23053/18A
31 May 1990
SUPERSEDING
MIL-I-23053/18
5 August 1988

MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, MODIFIED FLUOROPOLYMER, CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this Specification Sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-23053.

REQUIREMENTS:

Continuous operating temperature range: Class 1: -55°C (-67°F) to +175°C (347°F). Classes 2 and 3: -55°C (-67°F) to +150°C (+302°F).

Color: The sleeving shall be supplied in colors and clear. Colors shall conform to the requirements of Class 1 of MIL-STD-104.

Class: The sleeving shall be furnished in the following classes, as specified (see 6.2.1):

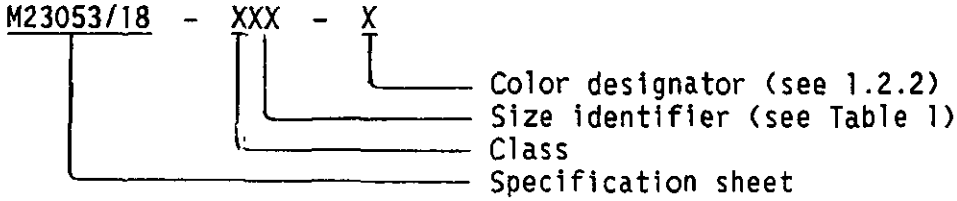
Class 1 - Semi-rigid. (Secant modulus over 100,000 psi (690(Mpa).

Class 2 - Flexible. (Secant modulus 25,000 to 120,000 psi (172 to 828 Mpa)

Class 3 - Very flexible. (Secant modulus to 25,000 psi (172 Mpa) Maximum.)

Longitudinal Change: +5, -10 percent; overexpanded sleeving: -15 percent.

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



Example: Class 1, Clear, 0.250 inch (6.35 mm) as supplied ID sleeving shall be identified as M23053/18-106-C.

AMSC N/A FSC 5970

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TABLE I. Construction details, inches (mm). 1/

Military part number 3/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/18-101-*	.046(1.17)	.023(.58)	.010 ± .002(.254 ± .051)
M23053/18-102-*	.063(1.60)	.031(.79)	.010 ± .002(.254 ± .051)
M23053/18-103-*	.093(2.36)	.046(1.17)	.010 ± .002(.305 ± .051)
M23053/18-104-*	.125(3.18)	.062(1.58)	.010 ± .002(.305 ± .051)
M23053/18-105-*	.187(4.75)	.093(2.36)	.010 ± .002(.305 ± .051)
M23053/18-106-*	.250(6.35)	.125(3.18)	.012 ± .003(.356 ± .076)
M23053/18-107-*	.375(9.53)	.187(4.75)	.012 ± .003(.356 ± .076)
M23053/18-108-*	.500(12.7)	.250(6.35)	.012 ± .003(.356 ± .076)
M23053/18-109-*	.750(19.1)	.375(9.53)	.017 ± .003(.432 ± .076)
M23053/18-110-*	1.000(25.4)	.500(12.7)	.019 ± .003(.483 ± .076)
M23053/18-111-*	1.500(38.1)	.750(19.1)	.020 ± .003(.508 ± .076)
M23053/18-112-*	2.000(50.8)	1.000(25.4)	.020 ± .003(.508 ± .076)
Class 2			
M23053/18-201-*	.046(1.17)	.023(.58)	.010 ± .002(.254 ± .051)
M23053/18-202-*	.063(1.60)	.031(.79)	.010 ± .002(.254 ± .051)
M23053/18-203-*	.093(2.36)	.046(1.17)	.010 ± .002(.305 ± .051)
M23053/18-204-*	.125(3.18)	.062(1.58)	.010 ± .002(.305 ± .051)
M23053/18-205-*	.187(4.75)	.093(2.36)	.010 ± .002(.305 ± .051)
M23053/18-206-*	.250(6.35)	.125(3.18)	.012 ± .003(.356 ± .076)
M23053/18-207-*	.375(9.53)	.187(4.75)	.012 ± .003(.356 ± .076)
M23053/18-208-*	.500(12.7)	.250(6.35)	.012 ± .003(.356 ± .076)
M23053/18-209-*	.750(19.1)	.375(9.53)	.017 ± .003(.432 ± .076)
M23053/18-210-*	1.000(25.4)	.500(12.7)	.019 ± .003(.483 ± .076)
OVEREXPANDED SLEEVING			
M23053/18-211-*	.093(2.36)	.031(.79)	.017 ± .003(.432 ± .076)
M23053/18-212-*	.187(4.75)	.062(1.58)	.020 ± .003(.508 ± .076)
M23053/18-213-*	.375(9.53)	.125(3.18)	.020 ± .003(.508 ± .076)
M23053/18-214-*	.750(19.1)	.250(6.35)	.025 ± .003(.635 ± .076)
M23053/18-215-*	1.000(25.4)	.333(8.46)	.030 ± .005(.762 ± .127)
Class 3			
M23053/18-301-*	.046(1.17)	.023(.58)	.010 ± .002(.254 ± .051)
M23053/18-302-*	.063(1.60)	.031(.79)	.010 ± .002(.254 ± .051)
M23053/18-303-*	.093(2.36)	.046(1.17)	.010 ± .002(.305 ± .051)
M23053/18-304-*	.125(3.18)	.062(1.58)	.010 ± .002(.305 ± .051)
M23053/18-305-*	.187(4.75)	.093(2.36)	.010 ± .002(.305 ± .051)
M23053/18-306-*	.250(6.35)	.125(3.18)	.012 ± .003(.356 ± .076)
M23053/18-307-*	.375(9.53)	.187(4.75)	.012 ± .003(.356 ± .076)
M23053/18-308-*	.500(12.7)	.250(6.35)	.012 ± .003(.356 ± .076)
M23053/18-309-*	.750(19.1)	.375(9.53)	.017 ± .003(.432 ± .076)
M23053/18-310-*	1.000(25.4)	.500(12.7)	.019 ± .003(.483 ± .076)

1/ Diameter limits for object to be enclosed shall be as recommended in technical data.
2/ Wall thickness values are less when shrinkage is restricted.
3/ The asterisk in the part number shall be replaced by color code designations (see 1.2.2).

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Unrestricted shrinkage: Test method 4.6.5: 200° ± 2°C (392° ± 4°F) for 3 minutes.

TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table 1	4.6.3
Heat shock	No cracks, flowing or dripping	4.6.8 275° ± 3°C (527° ± 6°F)
Secant modulus, psi (MPa)		4.6.13.1, 2 percent strain, ASTM D 882
Class 1	100,000 (690) min	
Class 2	25,000 (172.4) to 120,000 (828)	
Class 3	25,000 (172.4)max	
Restricted shrinkage	No cracks	4.6.6.1.1 200° ± 2°C (392° ± 4°F)
Voltage withstand	Pass	4.6.6.2
Clarity stability (Clear only)	Pass	4.6.17 200° ± 1°C (392° ± 2°F), 24 hours
Concentricity, min.	70% (50% for over expanded sizes)	4.6.3.3
Melting point, °C (°F), min.		ASTM D 4591 2/
Class 1	140 (284)	
Class 2	140 (284)	
Class 3	90 (194)	
After unrestricted shrinkage:		
ID, maximum	Table 1	4.6.3
Wall thickness	Table 1	4.6.3
Tensile strength, psi (MPa), minimum		4.6.14 ASTM D 638, 2 inches/min.
Class 1 & 2	3500 (24.3)	
Class 3	1500 (9.4)	

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TABLE II. Physical properties. (Continued) 1/

Characteristic	Requirement	Test procedure and conditions
Ultimate elongation, percent, minimum	Class 1 200 Class 2-300 Class 3-200	4.6.14 ASTM D 638, 2 inches/min.
Dielectric strength, volts/mil (Kv/mm), minimum	400(15.7)	4.6.2 ASTM D 2671
Volume resistivity, ohm-CM, minimum	10^{11}	4.6.2 ASTM D 876
Specific gravity, maximum	1.90	4.6.2 ASTM D 792
Water absorption, percent, maximum	0.5	4.6.2 ASTM D 570, 24 hrs. at 23°C
Corrosion	No corrosion	4.6.10.1 and 4.6.10.2 160° ± 2°C (320° ± 4°F), 16 hours
Low temperature flexibility	No cracking	4.6.7.1 -55° ± 1°C (-67° ± 2°F) 3/
Flammability	Pass	4.6.15.3
Heat resistance, property after:		4.6.9 Class 1: 250°±3°C(482°±5°F) for 168 hours. Classes 2 & 3: 225° ± 2°C (437° ± 4°F), for 168 hours
Ultimate elongation, percent, min.	Class 1-100 Class 2-250 Class 3-100	
Fluid resistance, property after:		4.6.11
Tensile strength, psi (MPa), minimum Class 1 & 2 Class 3	 2000 (13.9) 1000 (6.9)	
Elongation, percent, minimum	Class 1-100 Class 2-250 Class 3-100	
Dielectric strength, volt/mil (Kv/mm), minimum	400(15.7)	
Fungus resistance	Rating of 0	4.6.2 ASTM G 21

(Footnotes on next page)

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1/ Unless otherwise specified, requirements, test procedures, and conditions are for all sizes and colors of sleeving.
2/ Test procedure in accordance with ASTM D 4591. Melting point calculated as specified in "Melting endotherm peak" of ASTM D 3222.
3/ Low temperature flexibility: Specimens 0.187 and smaller shall be tested as whole sections after unrestricted shrinkage. Specimens larger than 0.187 inch (4.75mm) shall be a 0.250 inch (6.35mm) wide strip cut from the sleeving. All specimens shall be tested after unrestricted shrinkage. Mandrels shall be 10 times the specimen O.D. ± 10% for tubular specimens or 10 times the specimen thickness ± 10% for strip specimens.

Shelf life conditions: Supplier shall certify to storage at 18° to 35°C (65° to 95°F) for 4 years. Conformance to 3.5. See 3.5.2 for shelf life extension.

Intended use:

MIL-I-23053/18 sleevings are intended for use as wire markers, insulation and wire and component relief. Clear materials may be considered for use as replacements for non flame retarded clear sleevings.

Cross reference: Class 2 of revision A (MIL-I-23053/18A) is completely equivalent to the product contained in the superseded document (MIL-I-23053/18). Classes 1 and 3 are new.

Revision letters are not used to denote changes due to the extensiveness of changes.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - MI, MU
Navy - EC
Air Force - 80, 99

User activities:
Army - ME
Navy - MC

Preparing Activity
Navy - AS
Project Number 5970-1059

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